

Computer Science Major Course Planner with LLM Augmentation

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Introduction

- Students only have 4 years to complete all their courses (depth + breadth)
- Higher level courses have multiple prereqs and some courses only offered during certain quarters
- LLMs struggle with this type of reasoning

Materials and Methods

Data Acquisition:

Data from NU website about CS Degree Requirements, Course Information/Timings saved as tabular data.

Reasoning Engine

Architecture:

Knowledge Graph with Prerequisites in a Tree Structure so we can infer start and stop node for students.

Front-End:

Streamlit GUI from Python
Output is a graph of possible course plan

System Architecture

User Query Reasoning Engine

LLM

Identifies possible paths,
based on specialization and
prereqs

Natural Language Response
and Graph Output to User



Students input courses taken and desired, the tool suggests a feasible route with all prerequisites.

Conclusions

Students can use the tool to come up with a 4 year plan easily

Allows student to get specific course prerequisite information easily without having to search in multiple places

Future Steps

Include data about different courses which come under different schools but are still degree requirements

Expand to include different majors under McCormick

Github:

https://github.com/prakeerthprasadoff/Course_PreReq_Calculator



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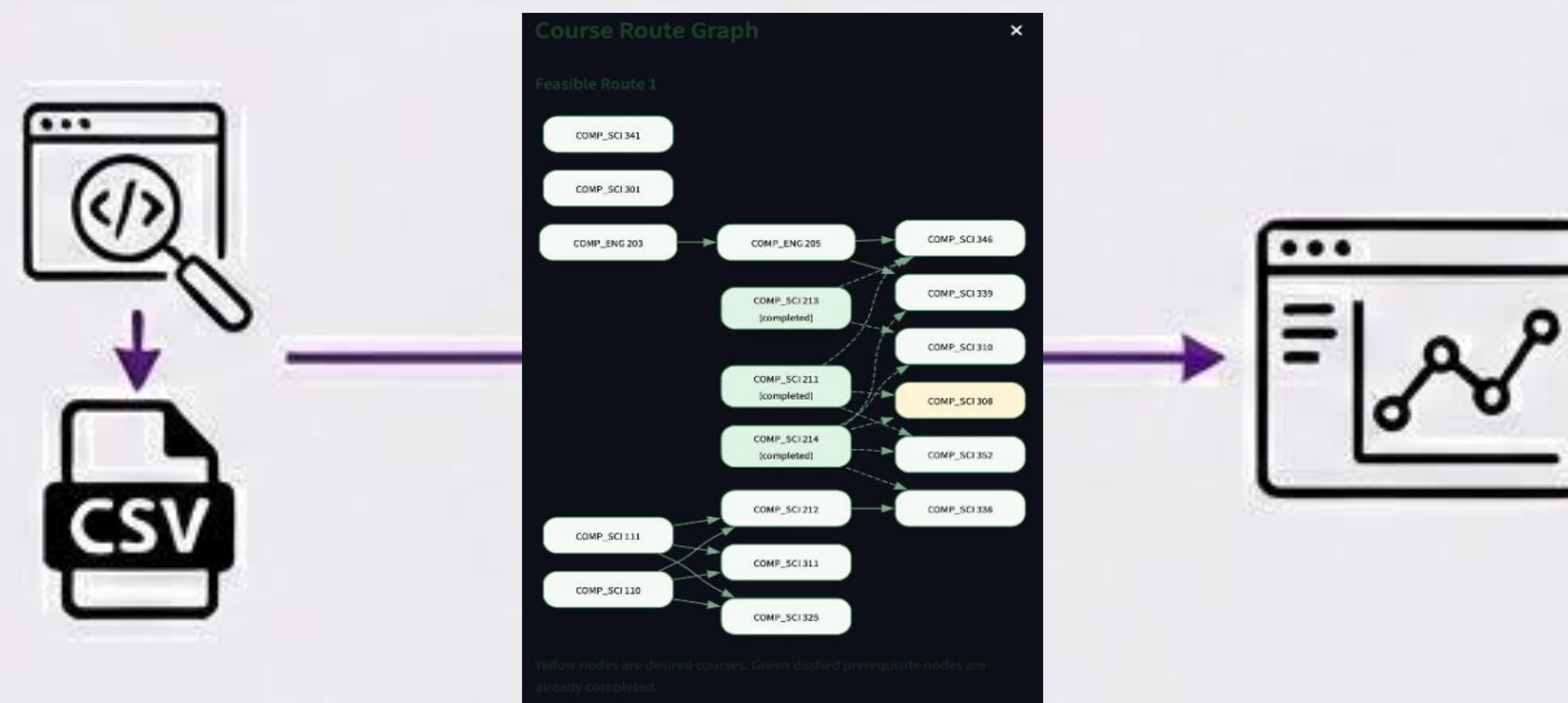
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Future Steps

- Include data about different courses which come under different schools but are still degree requirements
- Expand to include different majors under McCormick

Materials and Methods



Data Acquisition
Data from NU website about CS Degree Requirements, Course information/ Timings saved as CSV

Back-end Architecture
Knowledge Graph with Prerequisites in a Tree Structure so we can infer start and stop node for students. This was computed from the initial CSV.

Front-End
Streamlit GUI from Python. Output is a graph of possible course plan

Students input courses taken and desired, the tool suggests a feasible route with all prerequisites.

Conclusions & Information

Conclusions

- ✓ Students can use the tool to come up with a 4 year plan easily
- ✓ Allows student to get specific course prerequisite information easily without having to search in multiple places

Acknowledgments

-  We'd like to acknowledge Professor Alam for his assistance throughout this quarter. Department of Computer science, for allowing us to use their course catalog

Further information



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